Normalization source-fidelity review contract
T-1187; 2026-10-31.

Producer-organized preparation; independent verdict pending. Bound 13 packets, checked 902 included custody hashes, and matched 107 selected input files to Git index blobs at producer head `3388a75cf5`.

These checks bind bytes, not theorem truth or execution authenticity. No fresh proof replay or scientific execution occurred. Upstream compiled dependencies retain their trust boundary. Each role below is unassigned; independent reviewers must work in separate sessions and seal reports before seeing other findings.

1. Source identity

Claim: canonical choose/transform/map_divisor and related collector bodies

Attack: Try the known wrong first-61-line gate excerpt; demand exact named ranges and explicit identifier changes.

Blind re-derivation: Independently extract each named body and reconstruct call order.

2. Word arithmetic

Claim: small powers, binomial recurrence, wide modular product and inversion

Attack: Use p=0,1, composite p, noncanonical residues, a=0, k>5 and maximal words; identify excluded domains before applying lemmas.

Blind re-derivation: Derive each intermediate bound and inverse premise directly from Rust operations.

3. Traversal and storage

Claim: nested immutable coefficient / mutable slot traversal, trim and pad

Attack: Drop a row, change iteration order, alias coefficient with output, insert an internal zero, exceed capacity.

Blind re-derivation: Derive the coefficient sum and prove exact slice lengths from source degree constraints.

4. Representation premises

Claim: degree metadata, monicity, canonical inputs and ignored array tails

Attack: Supply degree metadata inconsistent with natDegree, degree>2, or nonzero unused tails; check what is assumed and what source actually validates.

Blind re-derivation: Trace constructors/callers that establish each premise; do not infer validity from array length.

5. Degree-zero branch

Claim: source returns original divisor unchanged; model proves decoded identity validity

Attack: Keep arbitrary unused storage words with degree0; distinguish byte identity from decoded polynomial identity.

Blind re-derivation: Derive u=1,v=0 from the precise validity predicate and degree premise.

6. Mumford predicate

Claim: monic u, degree bound, ordinate bound and divisibility

Attack: Use singular f, a=0 and nonmonic u; determine whether any geometric claim was smuggled into polynomial validity.

Blind re-derivation: Derive scaling powers d,3,6 and transport each of the four clauses.

7. Inverse and sign

Claim: polynomial round trips and affine signed-support equivalence

Attack: Reverse direction, negate beta incorrectly, omit y scaling, or discard sign; verify which controls detect each mutation.

Blind re-derivation: Derive inverse coordinate map from x_old=a*x_new+b and y_old=a^3*y_new.

8. Multiplicity

Claim: all-power divisibility and exact finite polynomial multiplicity

Attack: Use zero polynomial, k=0, repeated root, ramified curve point; reject any inference equating polynomial and curve valuation without a bridge.

Blind re-derivation: Derive quotient transport and both directions using the inverse; state valuation obligations separately.

9. Proof execution and dependencies

Claim: native checks, axiom audit, compiled cache trust boundary

Attack: Alter imported helper after receipt; omit a module; use prebuilt olean with mismatched source.

Blind re-derivation: Rebuild exact named modules in an isolated immutable tree and perform fresh axiom readback.

10. Geometric scope boundary

Claim: remaining local valuations, infinity and principal-divisor transport

Attack: Attempt to claim a Jacobian group isomorphism from rational-point tests or polynomial validity alone.

Blind re-derivation: Construct the curve/function-field map, extend it at infinity and derive local orders before group transport.

Required outputs: exact named-module rebuilds and fresh axiom audits in an isolated immutable input tree; clause-by-clause source mapping; explicit domains or counterexamples; independently authored scoped findings.

Runtime reconciliation: the saved mount-diff amendment remains unreviewed and unintegrated. The prior qualification remains failed. The campaign deadline remains 2026-10-31 06:00 UTC; no renewal, launch or reinterpretation of old results is made here.

Sources: review-evidence/references/lean-review.md; corrected canonical source map; preceding Mathlib affine-substitution references. Full paths and SHA256 bindings are in bindings.json.